

puter. Like you might expect, a VM functions just like an actual computer does. It creates its own CPU, Storage and Network Interface elements, which are software-based, this involves the creation of virtual servers on the physical server. Each virtual server runs an Operating System (OS), and applications running on the server view it as a standalone system. Thus, the OS can't differentiate between a virtual machine and a physical computer, effectively "believing" that the VM is a physical computer.

In the particular type of VMs that we'll be looking at, all the "guest" operating systems run on the same operating system kernel as the "host" system.

Let's simplify that statement a bit:

Imagine your "host" system is the Linux system you're running on your computer. Now, the guest systems are the other operating systems you are running on the Linux kernel. Simply put, the kernel is the core of your operating system. It is the base level software which has access to all parts of the machine.

The VM forms a virtual server and emulates the OS (and its corresponding applications) in that server. This emulation is done through what is known as a hypervisor, the "emulation engine".

The hypervisor maps the hardware aspects (CPU, hard drive etc) of the guest OS onto some physical space in the hardware of the computer. Hence, barring storage limitations, one can run several OSes this way. When you're booting your computer, you can choose which of the VMs you want to run. The VM then runs in a window, on the host operating system.

Most VMs fall into two categories:

- System Virtual Machine
This VM emulates all components and applications of a real computer.
- Process Virtual Machine
This VM is used to implement a single process. Its main use is to run applications incompatible with the host OS.

Benefits of a Virtual Machine

Decades ago, a big problem with using VMs in the industry was the noticeable difference in performance between virtual servers and actual, physical servers. Nowadays, with the tremendous exponential leaps that software has made, this performance gap is negligible, for the most part.

This has enabled industries to replace expensive physical servers running just one operating system with virtual machines running instances of various operating systems at once, hence significantly saving on costs and